Literature-to-Research Mapping (M1, M2, and M3)

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovels: Physics-Based External-Load Estimation, Payload Estimation and Machine Health Monitoring* **Researcher:** Suhail Majeed Sheikh, Junior Research Fellow, Department of Mining Engineering, IIT Kharagpur **Supervisor:** Prof. Sunita Mishra **Primary Sources Inspected:** papers_library.bib (58 BibTeX entries), papers.md (58 mining papers + 11 robotics estimation-theory additions), and M2_literature_findings.md.

1. Synthesis of the Literature Gap Driving M2 and M3

A systematic inspection of the 58-paper mining shovel library in papers_library.bib and M2_literature_findings.md establishes three foundational findings that define the scientific contribution of Milestones M2 and M3:

1. **Absence of Known-Truth External-Force Injection Validation (M2 Gap)**: No paper in the mining cable-shovel literature injects a known, time-varying external force F(t) into a verified multibody model and quantifies instantaneous generalized-force recovery Q_{\mathrm{ext}}(t) = J_s(q)^T F(t) against the floating-point subtraction floor. Existing studies either estimate a scalar post-dig payload mass during free-swing/hoist after bank exit (Rasuli et al., 2014), predict resistive forces from soil-mechanics equations without inverting motor/joint effort (Frimpong et al., 2005; Awuah-Offei & Frimpong, 2006; Stavropoulou et al., 2013), or train hybrid neural networks against empirical field logs where true bank contact force is unmeasured (Fu et al., 2022).
2. **Physical Origin of Through-Dig Non-Identifiability (M2 Publishable Core)**: While Palomba et al. (2019) documented severe ill-conditioning during through-dig joint-space state/load estimation on a hydraulic excavator and mitigated it numerically via Tikhonov regularization, M2 derives the exact closed-form physical mechanism on the cable-shovel saddle--crowd mechanism: the lever-arm ratio between an external vertical load at the bail (s_{\mathrm{bail}} = d_4 + 1.50\ \mathrm{m}) and dipper mass/station error at the center of gravity (r_{\mathrm{COG}} = d_4 - 1.32\ \mathrm{m}), \rho_{\mathrm{station}}(d_4) = \frac{d_4 + 1.50}{d_4 - 1.32}, \qquad \frac{\mathrm{d}\rho_{\mathrm{station}}}{\mathrm{d}d_4} = -\frac{2.82}{(d_4 - 1.32)^2}, collapses when the crowd stroke \Delta d_4 is restricted to a narrow operational band (\Delta d_4 = 0.54\ \mathrm{m} in Bi et al., 2020), causing external load and dipper mass error to become collinear (r = 0.99999193, subspace angle \theta_{\min} = 0.2302^\circ, condition number \kappa_2(\tilde{S}) = 497.71).
3. **Elimination of Numerical Acceleration Differentiation (M3 Momentum Observer Core)**: Stage M1 (validate_and_identify.m and Stage1_Report_M1_v3.docx) proved that replacing sensed joint accelerations (\ddot{q}_3, \ddot{d}_4) with numerical differentiation (gradient) of 30\ \mathrm{Hz} velocities biases the identified rotational inertia I_{zz} by +56.1\% (449{,}397\ \mathrm{kg\cdot m^2} vs. 287{,}900\ \mathrm{kg\cdot m^2}). Adopting the generalized-momentum observer framework of De Luca & Mattone (2003) and Haddadin et al. (2017) eliminates \ddot{q} completely via \dot{p} = \tau_a + \tau_L -

G(q) - F_v\dot{q} - F_c(\dot{q}) + C(q,\dot{q})^T\dot{q}.

2. Detailed Paper-by-Paper Mapping Table

Reference & Source Link	Physical System Studied	Governing Equations & Model Structure	Parameter Identification & Load Method	External Load & Digging Treatment	Measurements Required	Key Limitations	Direct Relation to M1, M2, and M3
Rasuli et al. (2014) (*IEEE Trans. Ind. Appl.*, 50(3)) & **Rasuli (2012)** PhD Thesis	P&H 2100BL electric mining cable shovel (M_d = 54{,}300\ \mathrm{kg}, I_{zz} = 287{,}900\ \mathrm{kg\cdot m^2}, L_{\mathrm{COG}} = 2.82\ \mathrm{m})	Planar 2-DOF Euler--Lagrange dipper--handle subsystem q = [q_3, d_4]^T with transmission kinematics mapping hoist rope and crowd motor to (q_3, d_4)	Linear-in-parameters ordinary least squares (OLS) \tau = Y(q,\dot{q},\ddot{q})\theta during free motion; algebraic payload mass inversion post-dig	Estimates **scalar payload mass** m_{\mathrm{pay}} only during free motion after bank exit; **no digging contact-force time series** F(t) recovered	Resolver angles/lengths (q_3, d_4) at 30\ \mathrm{Hz}, motor armature voltage and current	Assumes known/fixed payload centroid; requires post-dig free-swing window; does not estimate time-varying bank digging forces or solve through-dig non-identifiability	Provides the exact physical parameters locked in shovel_params.m (M1) and the 30\ \mathrm{Hz} sampling rate used in M3
Tafazoli et al. (1999) (*IEEE Trans. Robot. Autom.*, 15(5))	Takeuchi TB15 hydraulic mini-excavator arm (boom, stick, bucket)	Rigid-body Euler--Lagrange equations regrouped into base inertial parameter vector plus viscous and Coulomb friction	Filtered linear least squares avoiding direct double differentiation; separate sequential tests for gravity vs. inertial/friction parameters	Studies **payload mass** (lumped with bucket mass); no continuous external digging-force vector F(t) inversion	Joint angles (encoders) and hydraulic cylinder pressures	Sequential gravity-first identification assumes static/quasi-static poses; does not address continuous digging load vs. mass collinearity	Motivates M1 linear-in-parameters regressor Y\theta, M2 gravity-dominance analysis (40.25:1 to 42.71:1), and M3 acceleration-free momentum observer
Fu et al. (2022) (*Chin. J. Mech. Eng.*,	WK-series electric cable shovel front-end	2-DOF Lagrangian model with time-varying dipper	Hybrid physics loss (Lagrange + energy	Predicts dynamic digging load; demonstrat	Hoist/crowd displacements and motor drive power/curr	Empirical network component lacks closed-form	Directly motivates Revision **V22** and **V24** in

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35:147)	working mechanism	mass m_d(t) = m_0 + m_t(t) embedded in hybrid Physics-Informed Neural Network (HPINN)	conservation) + deep neural network trained on 50 cycles (30 synthetic + 20 field)	es in Eq. (3) that payload mass m_t enters **inertia, Coriolis, and gravity** terms simultaneously	ents	identifiability guarantees; field dataset has no independent ground-truth bank contact force sensor	M2_design_evolution.md: full mass sensitivity s_{M_d} = r^2\ddot{q}_3 + 2r\dot{d}_4\dot{q}_3 + gr\cos q_3 vs. point-force injection
Bi et al. (2020) (*Energies*, 13(16))	WK-55 electric cable shovel tested at Anjialing open-pit mine	Kinematic and dynamic digging trajectory model with multi-objective genetic algorithm (digging time vs. specific energy)	Constraint-based optimization using field-measured motor and geometric limits (d_4 \in [9.50, 10.04]\ \mathrm{m} during digging pass)	Models resistive force via empirical soil-cutting equations; does not invert external load from drive measurements	Field cycle time, crowd extension d_4, hoist rope length, and motor power	Operational constraints are specific to WK-55 (not P&H 2100BL); narrow crowd stroke (\Delta d_4 = 0.54\ \mathrm{m}) restricts lever-arm variation	Defines the **BI operational-band** trajectory (d_4 = 9.77 \pm 0.27\ \mathrm{m}) in M2 (04_frequency_sweep narrow family and M2_wrongstation_bi_band.mat)
Feng et al. (2022) (*Machines*, 10(12):1203)	1:30 scale WK-series electric cable shovel test bench + EDEM--RecurDyn	Bidirectional coupled Discrete Element Method (EDEM) + Multibody Dynamics (RecurDyn)	Direct forward co-simulation validated against physical scale-bench strain/force measurements	Computes and measures excavating resistance at the dipper teeth/bucket	Scale-bench force transducers and displacement sensors	Forward simulation only (computationally heavy EDEM); not an online inverse observer; scale distortions in granular media	Directly motivates Revision **V21** in recover_load.m: reporting peakMagErr_pct and peakTimeErr_s alongside NRMSE

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Raza & Frimpong (2017) & **Frimpong et al. (2005)**, **(2008)**	P&H 4100XPC / 4100C BOSS electric rope shovel front-end assembly	Newton--Euler and Lagrangian multibody dynamics coupled with ANSYS FEA fatigue/crack-propagation models	Forward dynamic simulation with analytical soil-formation resistive forces and migrating payload centroid	Models resistive force at dipper teeth and a **migrating payload centroid** as the dipper fills	Virtual prototype / FEA stress and kinematic histories	Forward stress/fatigue simulation; does not solve the inverse load-identification or station-confounding problem	Directly motivates Revision **V23** (M2_P06): testing fixed bail station (s = d_4 + 1.50\\mathrm{m}) vs. displaced station (s = d_4 - 1.32\\mathrm{m}) and documenting station-migration error
Palomba et al. (2019) (*Mech. Syst. Signal Process.*)	Hydraulic excavator arm during through-digging operations	Rigid-body inverse dynamics and joint-space load estimation	Identifies severe ill-conditioning in through-dig regression and applies numerical Tikhonov regularization	Estimates joint/external digging loads during active excavation	Joint angles and actuator pressures/forces	Treats ill-conditioning purely as a numerical regularization problem without isolating the geometric lever-arm ratio \rho(d_4)	Provides the primary benchmark contrast for M2: whereas Palomba regularizes an ill-conditioned matrix, M2 proves the physical cause (\Delta\rho(d_4) collapse) and evaluates excitation design

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Peralta et al. (2026) (*Int. J. Electr. Electron. Eng.*, 13(1):108) & **Valenzuela & Valenzuela (2014)** (*IEEE Trans. Ind. Appl.*)	AC electric rope shovel mechanical--electrical drive system	Coupled kinematic, dynamic, and electrical drive torque/speed/power cycle equations	Forward cycle evaluation of hoist and crowd drive torque, speed, and active power requirements	Evaluates digging and swing cycle power envelopes under prescribed load profiles	Motor electrical terminal quantities (voltage, current, power, speed)	Forward power/torque sizing analysis; does not formulate a closed-loop disturbance observer for unknown external bank loads	Links the mechanical generalized forces (\tau_3, f_4) estimated in M2/M3 to the electrical drive health monitoring objectives of parent CMPDI project IIT/SRIC/R/AEH/2026/104
De Luca & Mattone (2003) (*IEEE ICRA*) & **Haddadin et al. (2017)** (*IEEE Trans. Robot.*, 33(6))	Rigid-link robotic manipulators subject to actuator faults and external contact wrenches	Generalized momentum p = D(q)\dot{q} with identity \dot{D} = C + C^T, yielding \dot{p} = \tau_a + \tau_L - G - F_v\dot{q} - F_c + C^T\dot{q}	First-order stable momentum residual observer \dot{z} = h(q,\dot{q},\tau_a) + K_O(p - z), \hat{\tau}_L = K_O(p - z)	Estimates arbitrary time-varying external generalized load \tau_L(t) with first-order low-pass dynamics G_O(s) = K_O(sI + K_O)^{-1}	Joint positions q, velocities \dot{q}, and actuator efforts \tau_a (**zero acceleration \ddot{q} required**)	Assumes exact friction/inertia model in standard form; discrete-time implementation at low sampling rate (30\ \mathrm{Hz}) requires careful handling of stiff Coulomb transitions	Direct mathematical blueprint for **Milestone M3 (Lane B)**: Branch A (friction-aware RK4 substepped observer) and Branch B (friction-uncertainty exact FOH observer)
Swevers et al. (1997) (*IEEE Trans. Robot.*	Nonlinear multi-DOF robotic manipulators	Persistent-excitation regressor and sensitivity	Constrained trajectory optimization minimizing	Free-motion and loaded identification trajectory	Joint positions and velocities	Classical formulation optimizes free-motion base	Directly underpins the formal SVD conditionin

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Autom., 13(5)) & **Armstrong (1989)** (*IJRR*, 8(6))	undergoing system identification	matrices $Y(q,\dot{q},\ddot{q})$ / $S(q,\dot{q},\ddot{q})$	the condition number $\kappa(S)$ subject to joint position, velocity, and acceleration bounds	design		parameters rather than joint load-vs-mass separability	g analysis ($\kappa(S)$, singular values σ_i, and principal subspace angles $\theta_{\min}$) in M2_FORMAL_IDENTIFIABILITY_REPORT.md
Rill, Schaeffer & Schuderer (2023) (*Multibody Syst. Dyn.*)	Multibody mechanical systems with dry/Coulomb and bristle friction	Comparison of LuGre bristle dynamics vs. regularized static friction $F_c(\dot{q}) = f_c\tanh(\dot{q}/\epsilon)$	Analytical and numerical stiffness/drift evaluation during stick-slip and velocity reversals	Evaluates friction force fidelity across zero-velocity crossings	Position and velocity trajectories across reversals	High-slope regularized $\tanh(\dot{q}/\epsilon)$ ($\epsilon \sim 10^{-4}\text{--}10^{-3}$) is numerically stiff at coarse sample steps ($T_s = 1/30\ \mathrm{s}$)	Explains the stiffness ratio $L_{c,i}/k_i$ at velocity zero-crossings and justifies the Branch A vs. Branch B split in M3